

SingleRAN

VRF Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-09-26).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Extended the VRF-related parameter specifications for the UMPThb. For details, see 3.3.3 Hardware .	None	NR LTE	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to NR, LTE FDD, LTE TDD, and NB-IoT. This feature works the same way with each of these RATs.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE TDD	TDLBFD-00113 4	Virtual Routing and Forwarding	3 VRF

3 VRF

3.1 Principles

3.1.1 Basic Principles

Background

When multi-operator sharing or network slicing is used, service interface IP addresses need to be separately planned for each operator or slice. This may lead to conflicts. To avoid this problem, network design personnel can use the virtual routing and forwarding (VRF) function to create different VRF instances on the base station and map the service interface IP addresses of different operators or network slices to corresponding VRF instances. This prevents packet exchanges between networks of operators or slices of different VRF instances, thereby ensuring independent operation and secure isolation for the operational data of each operator or slice. VRF facilitates the creation of multiple virtually isolated, end-to-end (E2E) logical networks on one physical network, enabling operators or industry users to share devices.

Related Concepts

- VRF instance
The VRF index and user label on the base station side are specified by the **VRF.VRFIDX (LTE eNodeB, 5G gNodeB)** and **VRF.USERLABEL (LTE eNodeB, 5G gNodeB)** parameters in the **VRF** MO, respectively. The VRF index identifies a VRF instance on a base station. Once a VRF instance is created on a base station, so is a routing table. The VRF instance with the index of 0 functions as the default VRF instance and is implemented in the same way as other VRF instances. During data transmission, if no VRF instance index is specified, data will be transmitted through the default VRF instance.
- Interface
The base station uses the **INTERFACE** MO to manage both normal and VLAN interfaces. The interface type is specified by the **INTERFACE.ITFTYPE (LTE eNodeB, 5G gNodeB)** parameter. An interface is associated with a specified VRF instance, which is VRF index 0 by default.

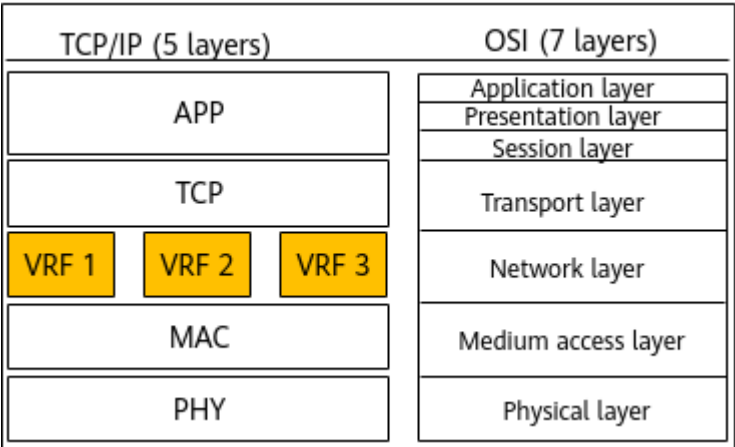
VRF can be used only when **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**.

Principles

The base station, in its transmit direction, determines the destination VRF instance based on the application layer protocol and indicates it using **VRF.VRFIDX (LTE eNodeB, 5G gNodeB)**.

In the receive direction of the base station, data is sent to the VLAN interface (specified by the **INTERFACE MO**) corresponding to the specified VRF index before being sent to the application layer. At the network layer, the base station uses VRF indexes to identify and isolate packets belonging to different logical networks. At the data link and physical layers, packets from different VRF instances are differentiated by VLAN ID or physical port. Therefore, when VRF is enabled on interfaces that adopt different VRF indexes (specified by **INTERFACE.VRFIDX (LTE eNodeB, 5G gNodeB)**), these interfaces must also have different VLAN IDs (specified by **INTERFACE.VLANID (LTE eNodeB, 5G gNodeB)**).

Figure 3-1 VRF position in the protocol architecture



3.1.2 Application Scenarios

VRF can be used in network sharing and network slicing scenarios. To use VRF, the VLAN function must be enabled on the transmission devices that are directly connected to the base station, and the VPN must be configured on routers. If transmission devices do not support VLANs or routers do not support VPNs, services of operators cannot be isolated. In addition, the transmission devices cannot work properly if there are IP address conflicts on the transport network.

Network Sharing Scenario

Figure 3-2 illustrates VRF networking in the network sharing scenario, and Table 3-1 describes the mapping relationships in the networking. When VRF is applied in the network sharing scenario, the VRF mapping relationships in the uplink of the base station are as follows: a public land mobile network (PLMN) is mapped to an operator, an operator is mapped to a VRF index, and a VRF index is mapped to a VLAN. In the downlink, the mapping relationships are reversed. In the network architecture, VRF is used on both the base station and VPN router, and services to

be isolated are transmitted in different VRF instances to their correct destinations (such as the core networks of different operators) based on the VRF mapping relationships.

Figure 3-2 VRF networking (network sharing scenario)

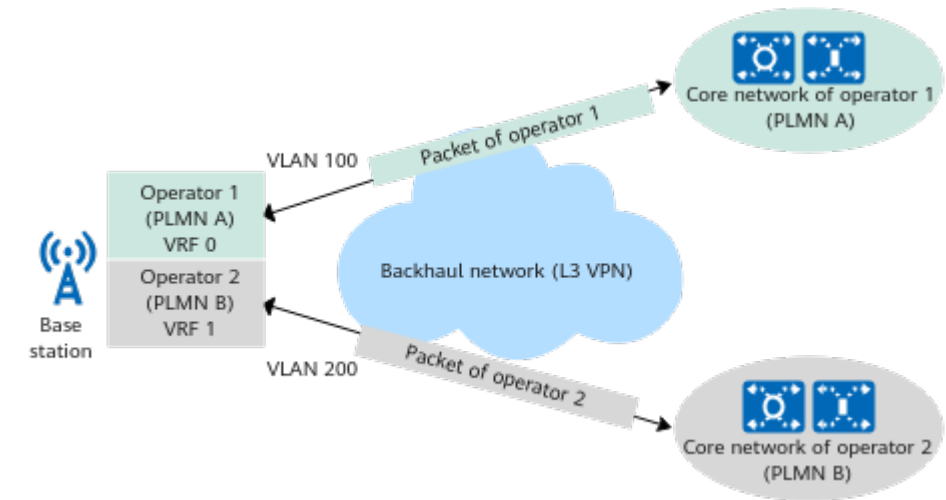


Table 3-1 Routing table

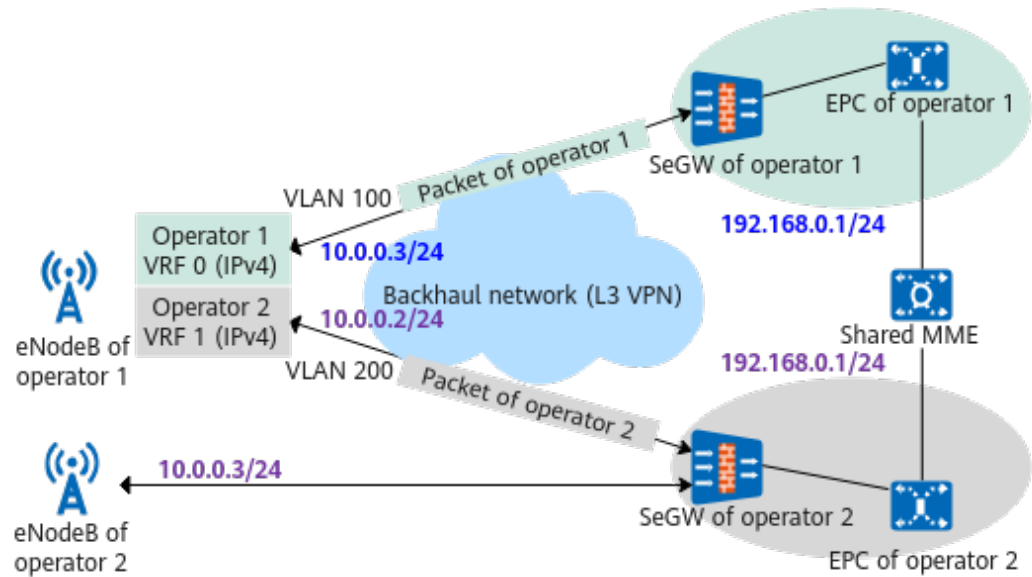
PLMN	Operator	VRF Index	VLAN
A	1	0	VLAN 100
B	2	1	VLAN 200

In the LTE TDD wholesale solution, there are three available network sharing architectures: multi-operator core network (MOCN), gateway core network (GWCN), and modified home routed (MHR). VRF can be used in MOCN and GWCN. For details about MOCN and GWCN, see *RAN Sharing* in *eRAN Feature Documentation*. For details about MHR, see *Wholesale (TDD)* in *eRAN Feature Documentation*.

In network sharing solutions for LTE TDD, LTE FDD, NB-IoT, and NR, VRF can be used in various architectures. For details, see *RAN Sharing* in *eRAN Feature Documentation* and *Multi-Operator Sharing* in *5G RAN Feature Documentation*.

For details about VRF deployment in the MOCN scenario, refer to the deployment in the IPv4 non-secure networking scenario. In this scenario, only VRF and VPN are configured on the base station and router, respectively. In the GWCN scenario, both the base station and MME are connected to two operators' networks, and VRF must be configured on the MME, as shown in [Figure 3-3](#).

Figure 3-3 Application of VRF in GWCN scenarios



Network Slicing Scenario

Figure 3-4 illustrates VRF networking in the network slicing scenario, and Table 3-2 describes the mapping relationships in the networking. When VRF is applied in the network slicing scenario, the VRF mapping relationships in the uplink of the base station are as follows: a Single Network Slice Selection Assistance Information (S-NSSAI) is mapped to a VRF index, and a VRF index is mapped to a VLAN. In the downlink, the mapping relationships are reversed. In the network architecture, VRF is used on both the base station and VPN router, and services to be isolated are transmitted in different VRF instances to their correct destinations (such as the AMF and UPF in different slices) based on the VRF mapping relationships.

Figure 3-4 VRF networking (network slicing scenario)

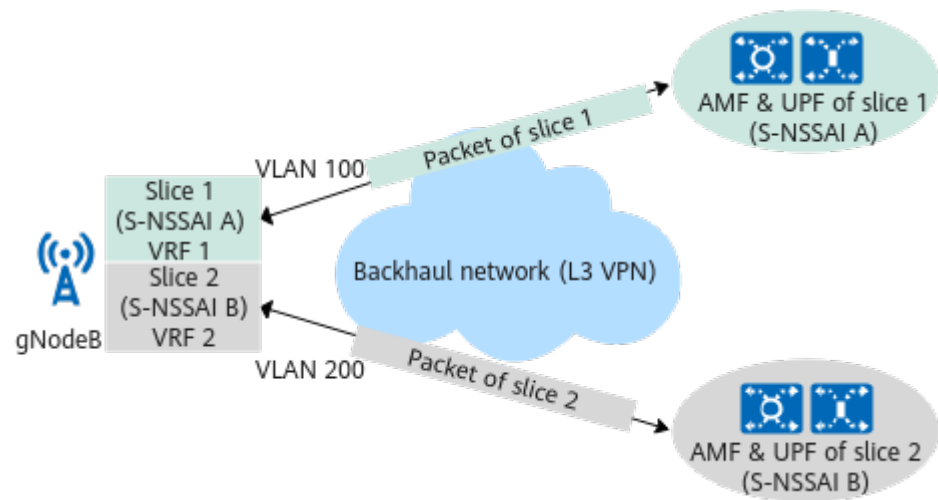


Table 3-2 Routing table

S-NSSAI	Slice	VRF Index	VLAN
A	1	1	VLAN 100
B	2	2	VLAN 200

 NOTE

- The mappings between operators' networks and VRF instances are as follows:
- If there are no network IP address conflicts among operators' networks, their networks can share the same default VRF instance.
 - If there are network IP address conflicts among operators' networks, their subnets must use different VRF instances.

3.1.3 Application Limitations

- The following limitations apply to the use of VRF:
- VRF instances do not support E1/T1 ports.
 - The DHCP relay, base station cascading, link transmission mode, and old transmission model can be deployed only in the default VRF instance.
 - In GUL co-MPT scenarios, GSM/UMTS services can be deployed only in the default VRF instance.
 - For IPv6 co-transmission of a separate-MPT MBTS, the IP addresses of the lower-level and upper-level base stations cannot be the same in different VRF instances.

3.1.4 Networking

3.1.4.1 Non-Secure Networking

Separate-MPT or Co-MPT Base Stations Using Independent Transmission

Base stations support VRF deployment in IPv4/IPv6 non-secure networking. VRF deployment in network sharing scenarios is used as an example. For details, see [Figure 3-2](#) and [Table 3-1](#).

Separate-MPT MBTSs Using Main-Control-Board-based Co-transmission Through Backplane Interconnection

VRF deployment is supported when a separate-MPT MBTS implements main-control-board-based IPv4/IPv6 co-transmission through backplane interconnection.

For the RAT that provides the co-transmission port, VRF deployment in network sharing scenarios is the same as that for [Separate-MPT or Co-MPT Base Stations Using Independent Transmission](#).

For the RAT that does not provide the co-transmission port, VRF deployment in network sharing scenarios requires that VRF of the RAT that does not provide the

co-transmission port be mapped to the RAT that provides the co-transmission port through the backplane tunnel (using the **BACKTUNNEL** MO). For details, see [Figure 3-5](#), [Table 3-3](#), and [Table 3-4](#).

Figure 3-5 VRF networking (in co-transmission through backplane interconnection scenarios)

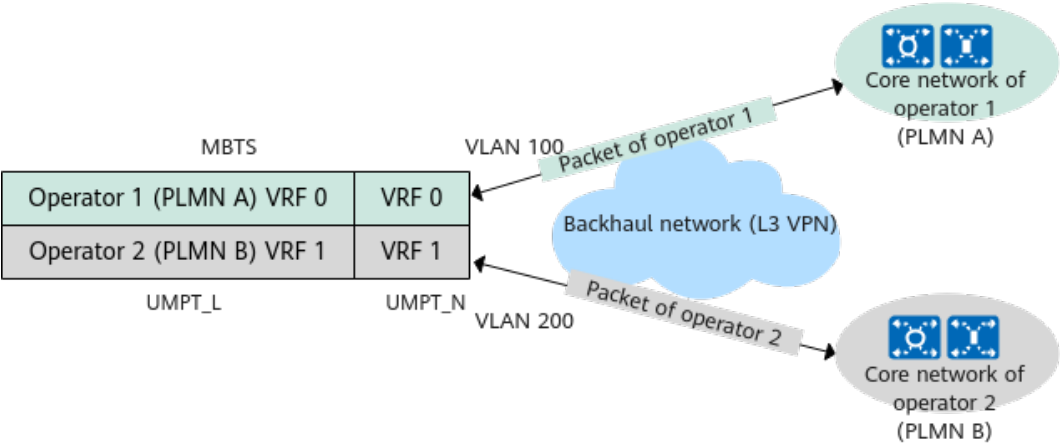


Table 3-3 Routing table (for the RAT that does not provide the co-transmission port)

PL M N	Oper ator	VRF Index	Backplane Tunnel
A	1	0	BACKTUNNEL.PORTID (LTE eNodeB, 5G gNodeB) is set to 0 , BACKTUNNEL.TUNNELTYPE (LTE eNodeB, 5G gNodeB) is set to NON_SHAPING , and BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) is set to 0 . (The tunnel number must be the same as that of the RAT that provides the co-transmission port.)
B	2	1	BACKTUNNEL.PORTID (LTE eNodeB, 5G gNodeB) is set to 1 , BACKTUNNEL.TUNNELTYPE (LTE eNodeB, 5G gNodeB) is set to NON_SHAPING , and BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB) is set to 1 . (The tunnel number must be the same as that of the RAT that provides the co-transmission port.)

Table 3-4 Routing table (for the RAT that provides the co-transmission port)

VRF Index	Backplane Tunnel	VLAN
0	BACKTUNNEL.PORTID (<i>LTE eNodeB, 5G gNodeB</i>) is set to 0 , BACKTUNNEL.TUNNELTYPE (<i>LTE eNodeB, 5G gNodeB</i>) is set to NON_SHAPING , and BACKTUNNEL.TUNNELNO (<i>LTE eNodeB, 5G gNodeB</i>) is set to 0 . (The tunnel number must be the same as that of the RAT that does not provide the co-transmission port.)	VLAN 100
1	BACKTUNNEL.PORTID (<i>LTE eNodeB, 5G gNodeB</i>) is set to 1 , BACKTUNNEL.TUNNELTYPE (<i>LTE eNodeB, 5G gNodeB</i>) is set to NON_SHAPING , and BACKTUNNEL.TUNNELNO (<i>LTE eNodeB, 5G gNodeB</i>) is set to 1 . (The tunnel number must be the same as that of the RAT that does not provide the co-transmission port.)	VLAN 200

 NOTE

For details about main-control-board-based co-transmission through backplane interconnection on a separate-MPT MBTS, see *Common Transmission*.

VRF deployment is only supported when the UMPT boards are used in co-transmission through backplane interconnection scenarios.

3.1.4.2 Secure Networking

Secure networking includes IPv4 networking and IPv6 networking. IPv6 networking includes IPv6-over-IPv6 secure networking and IPv4-over-IPv6 secure networking. In IPv6-over-IPv6 secure networking, IPv6-over-IPv6 IPsec is used. In IPv4-over-IPv6 secure networking, IPv4-over-IPv6 IPsec is used.

IPv6-over-IPv6 IPsec: IPv6 for both inner and outer IP headers

IPv4-over-IPv6 IPsec: IPv4 for the inner IP header and IPv6 for the outer IP header

3.1.4.2.1 IPv4 Secure Networking

This section uses the network sharing scenario as an example to describe the VRF deployment in IPv4 secure networking. VRF planning is related to the IPv4 IPsec configuration mode, which is specified by the **IKECFG.IPSECCFGMODE** (*5G gNodeB, LTE eNodeB*) parameter. When this parameter is set to **ETHINTERFACE_BASED**, inner service packets and outer IPsec packets must be configured in the same VRF instance, as shown in [Figure 3-6](#) and described in [Table 3-5](#). When this parameter is set to **TUNNELITF_BASED**, inner service packets and outer IPsec packets must be configured in different VRF instances, as shown in [Figure 3-7](#) and described in [Table 3-6](#).

To use VRF in IPv4 secure networking, **IKECFG.IPSECBINDMODE** (*LTE eNodeB, 5G gNodeB*) must be set to **SINGLE**. For IPv4 IPsec, the **IKEPEER.VRFIDX** (*LTE eNodeB, 5G gNodeB*) parameter specifies the VRF mapping.

Figure 3-6 VRF networking (Ethernet-interface-based IPv4 secure networking scenario)

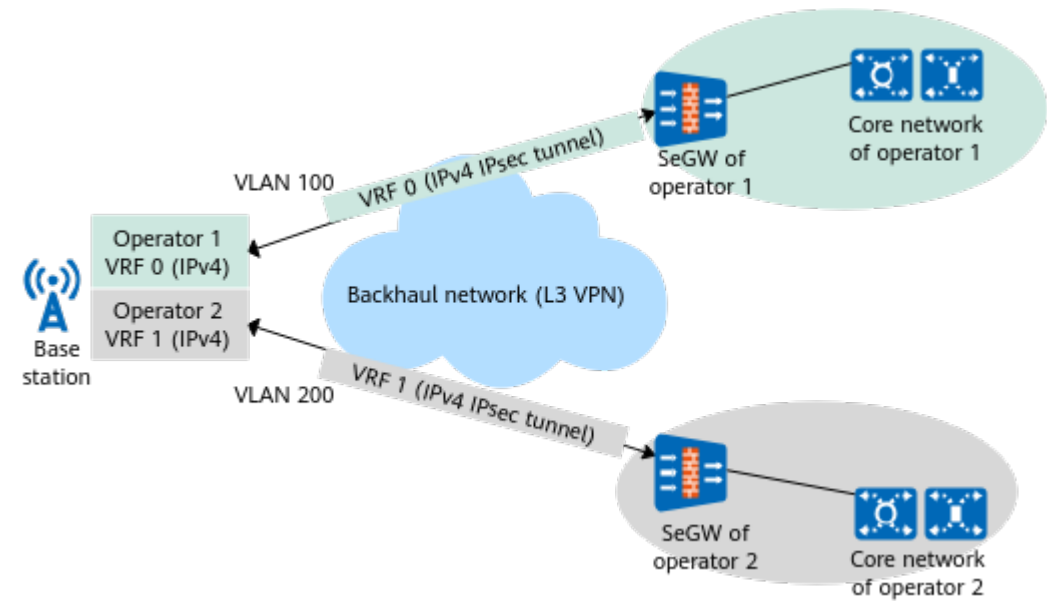


Table 3-5 Routing table

PLMN	Operator	VRF Index	VLAN
A	1	0 (for IPv4 services)	N/A
		0 (for IPv4 IPsec)	VLAN 100
B	2	1 (for IPv4 services)	N/A
		1 (for IPv4 IPsec)	VLAN 200

Figure 3-7 VRF networking (tunnel-interface-based IPv4 secure networking scenario)

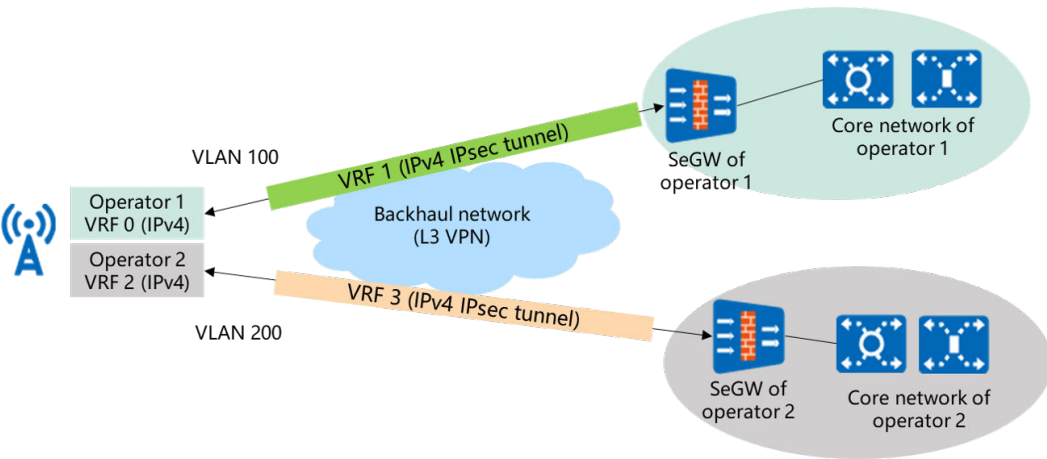


Table 3-6 Routing table

PLMN	Operator	VRF Index	VLAN
A	1	0 (for IPv4 services)	N/A
		1 (for IPv4 IPsec)	VLAN 100
B	2	2 (for IPv4 services)	N/A
		3 (for IPv4 IPsec)	VLAN 200

 NOTE

For details about IPsec principles, see *IPsec*.

When VRF is deployed in scenarios where a separate-MPT MBTS implements main-control-board-based IPv4 IPsec co-transmission through backplane interconnection, only inner IPsec services (plaintext that needs to be encrypted) are deployed at the RAT that does not provide the co-transmission port as IPsec encryption can only be deployed at the RAT that provides the co-transmission port. VRF deployment in backplane interconnection scenarios is the same as that described in "Non-Secure Networking." VRF deployment for the RAT that provides the co-transmission port is the same as that illustrated in [Figure 3-6](#) and [Table 3-5](#). For details about IPsec in co-transmission through backplane interconnection scenarios, see *IPsec*.

3.1.4.2.2 IPv6-over-IPv6 Secure Networking

This section uses the network sharing scenario as an example to describe the VRF deployment in IPv6-over-IPv6 secure networking scenarios. Unlike IPv4 secure networking, this scenario requires that inner service packets and outer IPsec packets be configured in different VRF instances. For details, see [Figure 3-8](#) and [Table 3-7](#). Inner and outer VRFs are configured to isolate inner IPsec service packets from outer IPsec service packets, preventing the inner service modules protected by IPsec from network attacks.

For IPv6 IPsec, the **TUNNELITE.VRFDX (LTE eNodeB, 5G gNodeB)** parameter specifies the VRF mapping.

Figure 3-8 VRF networking (IPv6-over-IPv6 secure networking scenario)

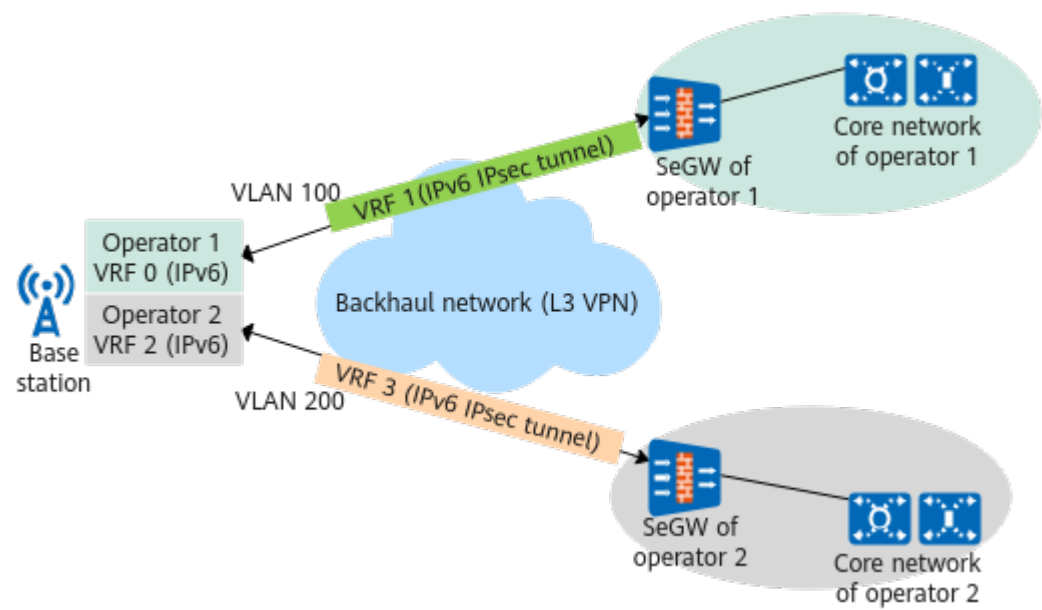


Table 3-7 Routing table

PLMN	Operator	VRF Index	VLAN
A	1	0 (for IPv6 services)	N/A
		1 (for IPv6 IPsec)	VLAN 100
B	2	2 (for IPv6 services)	N/A
		3 (for IPv6 IPsec)	VLAN 200

NOTE

VRF is used to isolate inner and outer packets in IPv6 secure networking. For details, see the description of VRF isolation in *IPsec*.

When VRF is deployed in scenarios where a separate-MPT MBTS implements main-control-board-based IPv6 IPsec co-transmission through backplane interconnection, only inner IPsec services (plaintext that needs to be encrypted) are deployed at the RAT that does not provide the co-transmission port as IPsec encryption can only be deployed at the RAT that provides the co-transmission port. VRF deployment in backplane interconnection scenarios is the same as that described in "Non-Secure Networking." VRF deployment for the RAT that provides the co-transmission port is the same as that illustrated in [Figure 3-8](#) and [Table 3-7](#). For details about IPsec in co-transmission through backplane interconnection scenarios, see *IPsec*.

3.1.4.2.3 IPv4-over-IPv6 Secure Networking

This section uses the network sharing scenario as an example to describe the VRF deployment in IPv4-over-IPv6 secure networking scenarios. In this case, the inner service packets and outer IPsec packets must be configured in different VRF instances. For details, see [Figure 3-9](#) and [Table 3-8](#).

Figure 3-9 VRF networking (IPv4-over-IPv6 secure networking scenario)

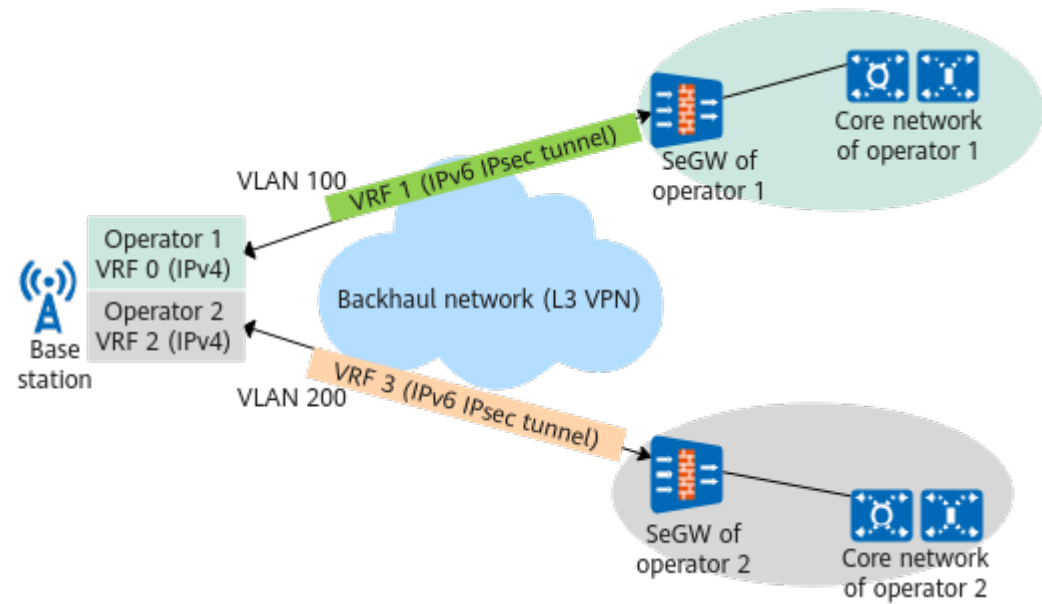


Table 3-8 Routing table

PLMN	Operator	VRF Index	VLAN
A	1	0 (for IPv4 services)	N/A
		1 (for IPv6 IPsec)	VLAN 100
B	2	2 (for IPv4 services)	N/A
		3 (for IPv6 IPsec)	VLAN 200

3.2 Network Analysis

3.2.1 Benefits

When multiple operators share a network or network slicing is used, service interface IP addresses need to be independently planned for each operator or slice. This may lead to conflicts. To prevent IP address conflicts within the same VRF instance, it is recommended that the VRF function be enabled to ensure that IP addresses do not conflict among operators' networks or slices.

3.2.2 Impacts

This function has no impact on the network.

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

3.3 Requirements

3.3.1 Licenses

There are no license requirements for basic functions.

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
VLAN support (IEEE 802.1p/q)	None	<i>IPv4 Transmission</i>	A physical port must be an Ethernet port or an Ethernet trunk. When VRF is enabled on interfaces that adopt different VRF indexes (specified by INTERFACE.VRFID (LTE eNodeB, 5G gNodeB)), these interfaces must have different VLAN IDs (specified by INTERFACE.VLANID (LTE eNodeB, 5G gNodeB)).

3.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
VLAN priority mapping	None	<i>IPv4 Transmission</i>	A non-default VRF instance (with the VRF.VRFID (LTE eNodeB, 5G gNodeB) parameter not set to 0) cannot coexist with a VLAN priority mapping (configured using the VLANCLASS MO).

Function Name	Function Switch	Reference	Description
Subnet VLAN mapping	None	None	A non-default VRF instance (with the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter not set to 0) cannot coexist with a subnet VLAN mapping (configured using the SUBNETVLAN MO).
PPP link	None	<i>IPv4 Transmission</i>	A non-default VRF instance (with the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter not set to 0) cannot coexist with a PPP link (configured using the PPPLNK MO).
MLPPP group	None	<i>IPv4 Transmission</i>	A non-default VRF instance (with the VRF.VRFIDX (LTE eNodeB, 5G gNodeB) parameter not set to 0) cannot coexist with an MLPPP group (configured using the MPGRP MO).

 NOTE

To deploy VRF in scenarios where a separate-MPT MBTS implements main-control-board-based IPv4/IPv6 co-transmission through backplane interconnection, the separate-MPT MBTS version must be SRAN18.1 or later.

3.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Direct IPv4 IPsec	None	<i>X2 and S1 Self-Management in NSA Networking S1 and X2 Self-Management NG and Xn Self-Management</i>	When the GTRANSPARA.TRANS CFGMO DE (LTE eNodeB, 5G gNodeB) parameter is set to OLD , direct IPv4 IPsec can be deployed only in the default VRF instance.
IPsec tunnel pair	None	<i>IPsec</i>	The IPsec tunnel pair can only be deployed in the default VRF instance.

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations
- DBS3900 LampSite and DBS5900 LampSite

Boards

Only the UMPT and UMDU boards support this function. When the UTRP or two main control boards are used, only the default VRF instance can be deployed. The following table lists the main control boards that can be used if the **VRF.VRFIDX (LTE eNodeB, 5G gNodeB)** parameter is set to a value greater than or equal to 8.

Main Control Board Type	Value of VRF.VRFIDX
UMPTg series	0 to 13
UMPTga series	
UMPTHa series	
UMPTHb series	0 to 31

RF Modules

This function does not depend on RF modules.

3.3.4 Networking

Before IP address planning, collect the IP address information for the core network NEs, MAE, and neighboring base stations. Ensure that there are no IP address conflicts within a VRF instance.

The VRF function can be used when the transmission links of the S1 and X2 interfaces are configured in the endpoint mode. For details, see *IP eRAN Engineering Guide* and *IP NR Engineering Guide*.

3.3.5 Others

None

3.4 Operation and Maintenance

3.4.1 Data Configuration

3.4.1.1 Data Preparation

Common Transmission Parameters

Related IPv4/IPv6 addresses and IPv4/IPv6 routes need to be configured for outer and inner VRF mapping. For details, see *IP eRAN Engineering Guide* in *eRAN Feature Documentation* and *IP NR Engineering Guide* in *5G RAN Feature Documentation*. The mapping relationships between IPv4/IPv6 routes and VRFs as well as between VRFs and backplane tunnels need to be configured. For details, see *Common Transmission*. The VRF mapping needs to be specified when interfaces, IP addresses, and IP routes are configured.

Networking-related Parameters

When a service link is configured in endpoint mode, the VRF mapping needs to be specified using the following parameters: **EPGROUP.VRFIDX (LTE eNodeB, 5G gNodeB)** (IPv4) or **EPGROUP.IPV6VRFIDX (LTE eNodeB, 5G gNodeB)** (IPv6), **SCTPHOST.VRFIDX (LTE eNodeB, 5G gNodeB)**, **SCTPPEER.VRFIDX (LTE eNodeB, 5G gNodeB)**, **USERPLANEHOST.VRFIDX (LTE eNodeB, 5G gNodeB)**, and **USERPLANEPEER.VRFIDX (LTE eNodeB, 5G gNodeB)**.

Parameters Related to Application Scenarios

- For details about service interface configuration for VRF in 5G network sharing scenarios, see *NG and Xn Self-Management* and *Multi-Operator Sharing* in *5G RAN Feature Documentation* as well as *X2 and S1 Self-Management in NSA Networking*.
- For details about service interface configuration for VRF in 5G network slicing scenarios, see *NG and Xn Self-Management* and *Network Slicing* in *5G RAN Feature Documentation*.
- For details about service interface configuration for VRF in 4G network sharing scenarios, see *RAN Sharing* and *S1 and X2 Self-Management* in *eRAN Feature Documentation*.

Basic Parameters

The parameters used for VRF activation when a separate-MPT or co-MPT base station uses independent transmission are as follows:

- VRF parameters: [Table 3-9](#) describes the parameters used for adding a VRF instance. The default VRF instance does not need to be added.

Table 3-9 VRF parameters

Parameter Name	Parameter ID	Setting Notes
VRF Index	VRF.VRFIDX (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan. You are advised to plan the VRF index for the entire network based on the operator or slice information.
User Label	VRF.USERLABEL (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.

- VLAN interface parameters: [Table 3-10](#) describes the parameters used for adding a VLAN interface to a physical port.

Table 3-10 VLAN interface parameters

Parameter Name	Parameter ID	Setting Notes
Interface ID	INTERFACE.ITFID (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
VRF Index	INTERFACE.VRFIDX (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
Interface Type	INTERFACE.ITFTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
Port Type	INTERFACE.PT (<i>LTE eNodeB, 5G gNodeB</i>)	For an Ethernet port, set this parameter to ETH . For an Ethernet trunk, set this parameter to ETHTRK .
Port ID	INTERFACE.PORTID (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
VLAN Tagged Switch	INTERFACE.TAGGED (<i>LTE eNodeB, 5G gNodeB</i>)	The default value is recommended.
VLAN ID	INTERFACE.VLANID (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
DSCP-to-PCP Mapping ID	INTERFACE.DSCP2PCP MAPID (LTE eNodeB, 5G gNodeB)	The default value is recommended.
DSCP-to-Queue Mapping ID	INTERFACE.DSCP2QUE MAPID (LTE eNodeB, 5G gNodeB)	The default value is recommended.
IPv4 Maximum Transmission Unit	INTERFACE.MTU4 (LTE eNodeB, 5G gNodeB)	The default value is recommended.
IPv6 Maximum Transmission Unit	INTERFACE.MTU6 (LTE eNodeB, 5G gNodeB)	The default value is recommended.
IPv6 Switch	INTERFACE.IPV6SW (LTE eNodeB, 5G gNodeB)	Set this parameter to ENABLE .
ARP Proxy	INTERFACE.ARPPROXY (LTE eNodeB, 5G gNodeB)	The default value is recommended.

- Normal interface (loopback port) parameters: [Table 3-11](#) describes the parameters used for adding a normal interface to a loopback port.

Table 3-11 Normal interface parameters

Parameter Name	Parameter ID	Setting Notes
Interface ID	INTERFACE.ITFID (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
VRF Index	INTERFACE.VRFIDX (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan to indicate the inner VRF. The value of this parameter must be different from that of the outer VRF. It is recommended that this parameter be set to 0 for the inner VRF of the primary operator.
Interface Type	INTERFACE.ITFTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to NORMAL .
Port Type	INTERFACE.PT (LTE eNodeB, 5G gNodeB)	Set this parameter to LOOPINT .

Parameter Name	Parameter ID	Setting Notes
Port ID	INTERFACE.PORTID (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

- IPsec parameters: For details about IPsec, see *IPsec*. This section focuses on the VRF mapping configuration in IPsec configurations.

For IPv4 IPsec, the **IKEPEER** MO is used to specify the VRF mapping. [Table 3-12](#) describes the parameters used for function activation.

Table 3-12 IKEPEER parameters

Parameter Name	Parameter ID	Setting Notes
IKE Peer Name	IKEPEER.PEERNAME (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
IP Version	IKEPEER.IPVERSION (LTE eNodeB, 5G gNodeB)	Set this parameter to IPV4 .
VRF Index	IKEPEER.VRFIDX (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Remote IP Address	IKEPEER.REMOTEIP (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Local IP Address	IKEPEER.LOCALIP (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

For IPv6 IPsec, the **TUNNELITF** MO is used to specify the VRF mapping. [Table 3-13](#) describes the parameters in the **TUNNELITF** MO.

Table 3-13 TUNNELITF parameters

Parameter Name	Parameter ID	Setting Notes
Mode	TUNNELITF.MODE (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Local IP Reference Method	TUNNELITF.LOCALIPREFMETHOD (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Local IPv6 Address	TUNNELITF.LOCALIP6 <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
Remote IPv6 Address	TUNNELITF.REMOTEIP6 <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
VRF Index	TUNNELITF.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.

- Direct IPsec parameters: For details about direct IPsec, see *S1 and X2 Self-Management*. This section focuses on the VRF mapping configuration in direct IPsec configurations.

For control-plane services, the **SCTPHOST** MO is used to specify the VRF mapping. [Table 3-14](#) describes the parameters in the **SCTPHOST** MO.

Table 3-14 SCTPHOST parameters

Parameter Name	Parameter ID	Setting Notes
VRF Index	SCTPHOST.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan. If IPSec Config Mode is set to ETHINTERFACE_BASED , the value of this parameter must be the same as that of SECURITYHOST.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i> , USERPLANEHOST.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i> , or EPGROUP.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i> (these three parameters share the same value). If IPSec Config Mode is set to TUNNELITF_BASED , the value of this parameter must be different from that of SECURITYHOST.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i> and the same as that of USERPLANEHOST.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i> or EPGROUP.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i> (these two parameters share the same value).
First Local IP Address	SCTPHOST.SIGIP1V4 <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
First Local IPv6 Address	SCTPHOST.SIGIP1V6 <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.

For user-plane services, the **USERPLANEHOST** MO is used to specify the VRF mapping. [Table 3-15](#) describes the parameters in the **USERPLANEHOST** MO.

Table 3-15 USERPLANEHOST parameters

Parameter Name	Parameter ID	Setting Notes
VRF Index	USERPLANEHOST.VRFIDX (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan. If IPSec Config Mode is set to ETHINTERFACE_BASED , the value of this parameter must be the same as that of SECURITYHOST.VRFIDX (LTE eNodeB, 5G gNodeB) , SCTPHOST.VRFIDX (LTE eNodeB, 5G gNodeB) , or EPGROUP.VRFIDX (LTE eNodeB, 5G gNodeB) (these three parameters share the same value). If IPSec Config Mode is set to TUNNELITF_BASED , the value of this parameter must be different from that of SECURITYHOST.VRFIDX (LTE eNodeB, 5G gNodeB) and the same as that of SCTPHOST.VRFIDX (LTE eNodeB, 5G gNodeB) or EPGROUP.VRFIDX (LTE eNodeB, 5G gNodeB) (these two parameters share the same value).
Local IP Address	USERPLANEHOST.LOCIPV4 (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Local IPv6 Address	USERPLANEHOST.LOCIPV6 (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

For the security host, the **SECURITYHOST** MO is used to specify the VRF mapping. [Table 3-16](#) describes the parameters in the **SECURITYHOST** MO.

Table 3-16 SECURITYHOST parameters

Parameter Name	Parameter ID	Setting Notes
VRF Index	SECURITYHOST.VRFID X (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan. If IPSec Config Mode is set to ETHINTERFACE_BASED , the value of this parameter must be the same as that of USERPLANEHOST.VRFID X (LTE eNodeB, 5G gNodeB) , SCTPHOST.VRFID X (LTE eNodeB, 5G gNodeB) , or EPGROUP.VRFID X (LTE eNodeB, 5G gNodeB) (these three parameters share the same value). If IPSec Config Mode is set to TUNNELITF_BASED , the value of this parameter must be different from that of any of USERPLANEHOST.VRFID X (LTE eNodeB, 5G gNodeB) , SCTPHOST.VRFID X (LTE eNodeB, 5G gNodeB) , and EPGROUP.VRFID X (LTE eNodeB, 5G gNodeB) .
Local IP Address	SECURITYHOST.LOCIKE IPV4 (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Local IPv6 Address	SECURITYHOST.LOCIKE IPV6 (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

- Backplane tunnel parameters: For VRF deployment in scenarios where a separate-MPT MBTS implements main-control-board-based IPv4/IPv6 co-transmission through backplane interconnection, the VRF parameters are the same as those listed above. For details about other parameter configurations, see *Common Transmission*. This section focuses on parameters related to the backplane tunnel.

Configure the backplane tunnel for the RAT that does not provide the co-transmission port. [Table 3-17](#) describes the parameters used for configuring the backplane tunnel.

Table 3-17 BACKTUNNEL parameters

Parameter Name	Parameter ID	Setting Notes
Source Cabinet No.	BACKTUNNEL.SCN (LTE eNodeB, 5G gNodeB)	Set this parameter to the cabinet number of the main control board of the RAT that does not provide the co-transmission port.
Source Subrack No.	BACKTUNNEL.SSRN (LTE eNodeB, 5G gNodeB)	Set this parameter to the subrack number of the main control board of the RAT that does not provide the co-transmission port.
Source Slot No.	BACKTUNNEL.SSN (LTE eNodeB, 5G gNodeB)	Set this parameter to the slot number of the main control board of the RAT that does not provide the co-transmission port.
Destination Cabinet No.	BACKTUNNEL.DCN (LTE eNodeB, 5G gNodeB)	Set this parameter to the cabinet number of the main control board of the RAT that provides the co-transmission port.
Destination Subrack No.	BACKTUNNEL.DSRN (LTE eNodeB, 5G gNodeB)	Set this parameter to the subrack number of the main control board of the RAT that provides the co-transmission port.
Destination Slot No.	BACKTUNNEL.DSN (LTE eNodeB, 5G gNodeB)	Set this parameter to the slot number of the main control board of the RAT that provides the co-transmission port.

Parameter Name	Parameter ID	Setting Notes
Tunnel Type	BACKTUNNEL.TUNNELTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to NON_SHAPING or SHAPING as required. If uplink traffic shaping is required, set this parameter to SHAPING . The main control board of the RAT that does not provide the co-transmission port supports only two backplane tunnels with the tunnel type of SHAPING .
Tunnel No.	BACKTUNNEL.TUNNELNO (LTE eNodeB, 5G gNodeB)	Different tunnel numbers are configured for backplane tunnels between the two main control boards based on the VRF mapping. The tunnel numbers configured for the two RATs must be the same in a VRF instance.

Configure normal interface (with **Port Type** set to **TUNNEL**) parameters for the RAT that does not provide the co-transmission port. [Table 3-18](#) describes the parameters used for adding a normal interface to a backplane tunnel.

Table 3-18 Normal interface parameters

Parameter Name	Parameter ID	Setting Notes
Interface ID	INTERFACE.ITFID (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
VRF Index	INTERFACE.VRFIDX (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan to indicate the VRF instance for services of the RAT that does not provide the co-transmission port.
Interface Type	INTERFACE.ITFTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to NORMAL .

Parameter Name	Parameter ID	Setting Notes
Port Type	INTERFACE.PT (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to TUNNEL .
Port ID	INTERFACE.PORTID (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to the value of the corresponding BACKTUNNEL.PORTID (<i>LTE eNodeB, 5G gNodeB</i>).

Configure the uplink route for the RAT that does not provide the co-transmission port. The traffic can only be diverted from the normal interface to the backplane tunnel. [Table 3-19](#) describes the parameters related to destination-based IPv4 routing as an example.

Table 3-19 Destination-based IPv4 routing parameters

Parameter Name	Parameter ID	Setting Notes
VRF Index	IPROUTE4.VRFIDX (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan to indicate the VRF instance for services of the RAT that does not provide the co-transmission port.
Destination IP	IPROUTE4.DSTIP (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
Mask	IPROUTE4.DSTMASK (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan.
Route Type	IPROUTE4.RTTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to IF .
Port Type	IPROUTE4.PT (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to INTERFACE .
Port ID	IPROUTE4.PORTID (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to the value of the corresponding INTERFACE.PORTID (<i>LTE eNodeB, 5G gNodeB</i>).

Configure the backplane tunnel for the RAT that provides the co-transmission port. [Table 3-20](#) describes the parameters used for configuring the backplane tunnel.

Table 3-20 BACKTUNNEL parameters

Parameter Name	Parameter ID	Setting Notes
Source Cabinet No.	BACKTUNNEL.SCN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the cabinet number of the main control board of the RAT that provides the co-transmission port.
Source Subrack No.	BACKTUNNEL.SSRN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the subrack number of the main control board of the RAT that provides the co-transmission port.
Source Slot No.	BACKTUNNEL.SSN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the slot number of the main control board of the RAT that provides the co-transmission port.
Destination Cabinet No.	BACKTUNNEL.DCN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the cabinet number of the main control board of the RAT that does not provide the co-transmission port.
Destination Subrack No.	BACKTUNNEL.DSRN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the subrack number of the main control board of the RAT that does not provide the co-transmission port.
Destination Slot No.	BACKTUNNEL.DSN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the slot number of the main control board of the RAT that does not provide the co-transmission port.
Tunnel Type	BACKTUNNEL.TUNNEL TYPE <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to NON_SHAPING .

Parameter Name	Parameter ID	Setting Notes
Tunnel No.	BACKTUNNEL.TUNNEL NO (LTE eNodeB, 5G gNodeB)	Different tunnel numbers are configured for backplane tunnels between the two main control boards based on the VRF mapping. The tunnel numbers configured for the two RATs must be the same.

Configure normal interface (with **Port Type** set to **TUNNEL**) parameters for the RAT that provides the co-transmission port. [Table 3-21](#) describes the parameters used for adding a normal interface to a backplane tunnel.

Table 3-21 Normal interface parameters

Parameter Name	Parameter ID	Setting Notes
Interface ID	INTERFACE.ITFID (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
VRF Index	INTERFACE.VRFIDX (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan to indicate the VRF instance for services of the RAT that does not provide the co-transmission port.
Interface Type	INTERFACE.ITFTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to NORMAL .
Port Type	INTERFACE.PT (LTE eNodeB, 5G gNodeB)	Set this parameter to TUNNEL .
Port ID	INTERFACE.PORTID (LTE eNodeB, 5G gNodeB)	Set this parameter to the value of the corresponding BACKTUNNEL.PORTID (LTE eNodeB, 5G gNodeB) .

Configure the route from the RAT that provides the co-transmission port to the RAT that does not provide the co-transmission port. The traffic can only be diverted from the normal interface to the backplane tunnel. [Table 3-22](#) describes the parameters related to destination-based IPv4 routing as an example.

Table 3-22 Destination-based IPv4 routing parameters

Parameter Name	Parameter ID	Setting Notes
VRF Index	IPROUTE4.VRFIDX <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan to indicate the VRF instance for services of the RAT that does not provide the co-transmission port.
Destination IP	IPROUTE4.DSTIP <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
Mask	IPROUTE4.DSTMASK <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the network plan.
Route Type	IPROUTE4.RTTYPE <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to IF .
Port Type	IPROUTE4.PT <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to INTERFACE .
Port ID	IPROUTE4.PORTID <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter to the value of the corresponding INTERFACE.PORTID <i>(LTE eNodeB, 5G gNodeB)</i> .

3.4.1.2 Using MML Commands

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Using network sharing in IPv6 non-secure networking as an example
//Configuring a VRF instance
ADD VRF: VRFIDX=1, USERLABEL="operator 2 VRF";

//Configuring VLAN interface parameters (using the Ethernet port as an example)
SET ETHPORT: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=0, PORTID=0;
ADD INTERFACE: ITFID=2, ITFTYPE=VLAN, PT=ETH, PORTID=0, VLANID=10, VRFIDX=1, IPV6SW=ENABLE,
```

```
DSCP2PCPMAPID=0, DSCP2QUEMAPID=0, MTU4=1500, MTU6=1500, ARPPROXY=DISABLE,
USERLABEL="operator 2 VLAN interface";

//Configuring parameters for a normal interface (loopback port)
ADD LOOPBACK: PORTID=0, SN=7;
ADD INTERFACE: ITFID=3, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=0, VRFID=1, USERLABEL="operator 2
Loopback interface";

//Using backplane tunnel configurations in IPv6 co-transmission through backplane interconnection as an
example
//Configuring a VRF instance for the RAT that does not provide the co-transmission port
ADD VRF: VRFID=1, USERLABEL="operator 2 VRF";

//Configuring backplane tunnel parameters for the RAT that does not provide the co-transmission port
ADD BACKTUNNEL: PORTID=1, SSN=6, DSN=7, TUNNELTYPE=NON_SHAPING, TUNNELNO=1;
//Configuring normal interface (with Port Type set to TUNNEL) parameters for the RAT that does not
provide the co-transmission port
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=TUNNEL, PORTID=1, VRFID=1, IPV6SW=ENABLE;
//Configuring destination-based routing parameters for the RAT that does not provide the co-transmission
port
ADD IPRROUTE6: RTID=1, VRFID=1, DSTIP="2001:db8:100:ad1:10:60::", PFXLEN=120, RTTYPE=IF, ITFID=1,
FORCEEXECUTE=NO;

//Configuring a VRF instance for the RAT that provides the co-transmission port
ADD VRF: VRFID=1, USERLABEL="backtunnel operator 2 VRF";

//Configuring backplane tunnel parameters for the RAT that provides the co-transmission port
ADD BACKTUNNEL: PORTID=1, SSN=7, DSN=6, TUNNELTYPE=NON_SHAPING, TUNNELNO=1;
//Configuring normal interface (with Port Type set to TUNNEL) parameters for the RAT that provides the
co-transmission port
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=TUNNEL, PORTID=1, VRFID=1, IPV6SW=ENABLE;
//Configuring destination-based routing parameters for the RAT that provides the co-transmission port
ADD IPRROUTE6: RTID=1, VRFID=1, DSTIP="2001:db8:100:ad1:10:20::", PFXLEN=120, RTTYPE=IF, ITFID=1,
FORCEEXECUTE=NO;

//Using network sharing in IPv6 secure networking as an example
//Configuring VRF instances
ADD VRF: VRFID=2, USERLABEL="operator 2 inner VRF";
ADD VRF: VRFID=3, USERLABEL="operator 2 outer VRF";

//Configuring VLAN interface parameters (using the Ethernet port as an example)
SET ETHPORT: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=0, PORTID=0;
ADD INTERFACE: ITFID=2, ITFTYPE=VLAN, PT=ETH, PORTID=0, VLANID=10, VRFID=3, IPV6SW=ENABLE,
DSCP2PCPMAPID=0, DSCP2QUEMAPID=0, MTU4=1500, MTU6=1500, ARPPROXY=DISABLE,
USERLABEL="operator 2 outer interface";

//Configuring parameters for a normal interface (loopback port)
ADD LOOPBACK: PORTID=0, SN=7;
ADD INTERFACE: ITFID=3, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=0, VRFID=2, USERLABEL="operator 2
inner interface";

//Configuring tunnel interface parameters
ADD TUNNELITF: PORTID=0, MODE=IPSEC_IPV6, LOCALIPREFMETHOD=IPADDR, VRFID=3,
LOCALIP6="2002::12:0:1", REMOTEIP6="2002::12:0:2";
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Removing VRF instances
RMV VRF: VRFID=2;
RMV VRF: VRFID=3;
```

3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.2 Activation Verification

Different VRF instances follow the same activation verification operations, except for configuring different VRF instance indexes. For details, see section "Deployment of Common Transmission Data" of *IP eRAN Engineering Guide* in *eRAN Feature Documentation* and section "Deployment of Common Transmission Data" of *IP NR Engineering Guide* in *5G RAN Feature Documentation*.

3.4.3 Network Monitoring

Table 3-23 lists the counters that can be used to observe the network-layer traffic of services of each operator or slice in each VRF instance and VLAN.

Table 3-23 Interface-related performance counters on the base station

Counter ID	Counter Name
1542460734	VS.INTERFACE.IPv4TxBytes (LTE eNodeB, 5G gNodeB)
1542460735	VS.INTERFACE.IPv4RxBytes (LTE eNodeB, 5G gNodeB)
1542460736	VS.INTERFACE.IPv6TxBytes (LTE eNodeB, 5G gNodeB)
1542460737	VS.INTERFACE.IPv6RxBytes (LTE eNodeB, 5G gNodeB)
1542460738	VS.INTERFACE.IPv4TxPackets (LTE eNodeB, 5G gNodeB)
1542460739	VS.INTERFACE.IPv4RxPackets (LTE eNodeB, 5G gNodeB)
1542460740	VS.INTERFACE.IPv6TxPackets (LTE eNodeB, 5G gNodeB)
1542460741	VS.INTERFACE.IPv6RxPackets (LTE eNodeB, 5G gNodeB)
1542460742	VS.INTERFACE.IPv4TxMaxSpeed (LTE eNodeB, 5G gNodeB)
1542460744	VS.INTERFACE.IPv4TxMeanSpeed (LTE eNodeB, 5G gNodeB)
1542460745	VS.INTERFACE.IPv4TxMinSpeed (LTE eNodeB, 5G gNodeB)
1542460746	VS.INTERFACE.IPv4RxMaxSpeed (LTE eNodeB, 5G gNodeB)

Counter ID	Counter Name
1542460747	VS.INTERFACE.IPv4RxMeanSpeed (LTE eNodeB, 5G gNodeB)
1542460748	VS.INTERFACE.IPv4RxMinSpeed (LTE eNodeB, 5G gNodeB)
1542460749	VS.INTERFACE.IPv6TxMaxSpeed (LTE eNodeB, 5G gNodeB)
1542460750	VS.INTERFACE.IPv6TxMeanSpeed (LTE eNodeB, 5G gNodeB)
1542460751	VS.INTERFACE.IPv6TxMinSpeed (LTE eNodeB, 5G gNodeB)
1542460752	VS.INTERFACE.IPv6RxMaxSpeed (LTE eNodeB, 5G gNodeB)
1542460753	VS.INTERFACE.IPv6RxMeanSpeed (LTE eNodeB, 5G gNodeB)
1542460754	VS.INTERFACE.IPv6RxMinSpeed (LTE eNodeB, 5G gNodeB)

For details about other counters related to network monitoring for this function, see *IP eRAN Engineering Guide* and *IP NR Engineering Guide*.

4 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- *IPsec*
- *IPv4 Transmission*
- *IPv6 Transmission*
- *Common Transmission*
- *X2 and S1 Self-Management in NSA Networking*
- Feature parameter description documents in *eRAN Feature Documentation*:
 - *IP eRAN Engineering Guide*
 - *Wholesale (TDD)*
 - *S1 and X2 Self-Management*
 - *RAN Sharing*
- Feature parameter description documents in *5G RAN Feature Documentation*:
 - *IP NR Engineering Guide*
 - *NG and Xn Self-Management*
 - *Multi-Operator Sharing*
 - *Network Slicing*